

TOPIC FIVE- FIRE PREVENTION AND CONTROL SERVICES

Introduction.

This is a combination of installations in a building which are meant for preventing and or stopping fire from increasing.

It also refers to **all systems, equipment, design strategies, and regulations** put in place within a building to:

- Prevent the outbreak of fire
- Detect fire early
- Control and contain fire spread
- Facilitate safe evacuation
- Enable firefighting operations

Fire prevention and control in Kenya are governed by the [Occupational Safety and Health Act \(OSHA\), 2007](#), and the **Fire Risk Reduction Rules of 2007**.

Fire Risk Reduction Rules, 2007: These rules require employers to maintain fire-fighting equipment, provide clearly marked exits, establish safe assembly points, and train personnel on fire safety. Neglecting fire safety regulations can lead to substantial fines, often up to KES 500,000.

Mandatory measures for workplaces include installing fire detection/suppression systems, conducting annual safety audits, training employees on emergency responses like the [RACE protocol](#) (Rescue, Alarm, Contain, Evacuate), and conducting fire drills at least once every 12 months.

Causes of fire in a building ?

Many fires in Kenya (e.g., markets like Gikomba, industries in Industrial Area, informal settlements like Kibera) are worsened by:

- Poor electrical wiring
- Overcrowding
- Lack of fire equipment

- Limited access roads for fire engines

Classification of Fire Fighting Systems and Equipment in Buildings.

1. **Passive controls**-The fire behaviour of a structure will depend on the nature of the materials used in construction. The aim of passive controls is to reduce the use of combustible materials to a minimum.

These are **built-in safety features** that do not require activation. **Examples:**

- Fire-resistant walls (e.g., concrete, masonry)
- Fire-rated doors (common in banks, hospitals)
- Compartmentation in malls like **Two Rivers Mall**

Functions:

- Contain fire within a section
 - Delay structural collapse
 - Protect escape routes
2. **Active Controls**-this are equipment installed in our buildings to fight the fire when it occurs. These are **systems that respond when fire occurs**. **Examples:**
 - Fire extinguishers
 - Hose reels
 - Sprinklers
 - Fire alarms

Active control can be achieved in many ways. The selection of the actual system depends on:

- the nature of the fire to be dealt with;
- the fire load (the combustible content per m² of floor area) (kJ/m²);
- the fire classification;
- the active involvement of the occupants in extinguishing the fire;
- the ability of the occupants to leave the building unaided i.e., children

Active control methods can be subdivided into first aid measures, involving the occupants using portable equipment to extinguish small fires, and fixed fire-fighting appliances.

Fire prevention and control services can provide a number of benefits, including:

- **Reduced risk of fires:** By eliminating or reducing fire hazards, fire prevention services can help to prevent fires from starting in the first place.

- **Early detection of fires:** Fire detection systems can alert people to a fire quickly, so that they can evacuate the building and call the fire department.
- **Faster extinguishment of fires:** Firefighting services can help to extinguish fires quickly and effectively, which can help to prevent major damage to property and loss of life.
- **Increased safety:** Fire prevention and control services can help to create a safer environment for everyone.

CLASSES OF FIRE.

Here's a breakdown of the main fire classes you might encounter in Kenya:

a) Class A: Ordinary Combustibles

These fires involve everyday materials like wood, paper, textiles, and rubber. Think of a cozy bonfire gone astray or a pile of leaves catching a spark.

Common example: Timber yards in Gikomba Market, Toi market

b) Class B: Flammable Liquids

Gasoline, kerosene, cooking oil, and paints are the usual suspects in these fires. Imagine a pan fire sizzling out of control or a fuel spill igniting.

Example: Petrol station

c) Class C: Electrical Equipment

Faulty wiring, overloaded circuits, and malfunctioning appliances can spark these electrically charged infernos. Think of sparks erupting from a power outlet or a short circuit in a machine.

d) Class D: Combustible Metals

Magnesium, titanium, and sodium are some of the metals that burn with intense heat, requiring specialized extinguishing agents. Think of Olympic rings spontaneously combusting.

Example: Industrial plant, rare but possible

e) Class K: Kitchen Fires

Cooking oils and fats deserve their own category, as they throw unique challenges at firefighters. Think of a deep fryer erupting in a fiery fountain.

Example: Restaurants, hotels, street food vendors

Other Acts or regulatory bodies that ensure safety in work place

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1. Fire Prevention Measures

- **Electrical Maintenance:** Regular inspections of wiring to avoid overloading sockets.
- **Storage of Flammables:** Proper storage of fuels and chemicals in well-ventilated, marked areas.
- **Inspections & Audits:** Routine Fire Safety Audits and Risk Assessments are mandatory to identify hazards.

2. Fire Protection & Control Services

One must make sure there is easily accessible firefighting equipment

- **Equipment Installation:** Mandatory presence of fire extinguishers (CO₂, Foam, Water), smoke detectors, fire alarms, and fire blankets.
- **Maintenance:** Routine inspection, servicing, and testing of extinguishers (monthly checks) and sprinkler systems to ensure functionality.
- **Emergency Response:** Establishment of fire teams, fire marshals, and clear evacuation procedures.

Where the substance cannot be eliminated, the judgement relating to what is reasonably practical must be informed by the risk assessment and measures must be introduced that:

- Control the risk at source
- Mitigate the effects of a fire
- Ensure the safe handling, storage, transportation of dangerous substances
- Maintain the necessary measures.

3. Training & Drills

- **Fire Drills:** Statutory Requirement: Must be conducted at least once every 12 months.

- **Training:** Employees must be trained on using fire extinguishers, identifying hazards, and following emergency procedures.
- **Acronym for Drills: RACE** (Rescue, Alarm, Contain, Evacuate) and **PASS** (Pull, Aim, Squeeze, Sweep) for extinguisher use.

Emergency routes and exit

Emergency routes and exits must:

- Emergency routes and exits must lead as directly as possible to a place of safety
- It must be possible to evacuate the premises quickly and safely
- The number, size and distribution of exits must be adequate for the maximum numbers of persons who may be present
- Emergency doors must open in the direction of escape
- Sliding/revolving doors must not be used as emergency exits
- Exit doors must be easily and immediately opened by any person who may need to use them in an emergency
- Escape routes must be indicated by signs
- Emergency routes and exits must be adequately illuminated in the event of a failure of the normal lighting.

General Fire Fighting Equipment

The following Firefighting systems and equipment are discussed in this unit:

- Portable Fire extinguishers
- Fire hose reels
- Automatic Sprinkler Systems
- Fire Hydrant.
- Drenchers
- Fire Dampers in Ventilation Ductwork
- Pressurization of Escape Routes
- Fire Detection and Alarms System.

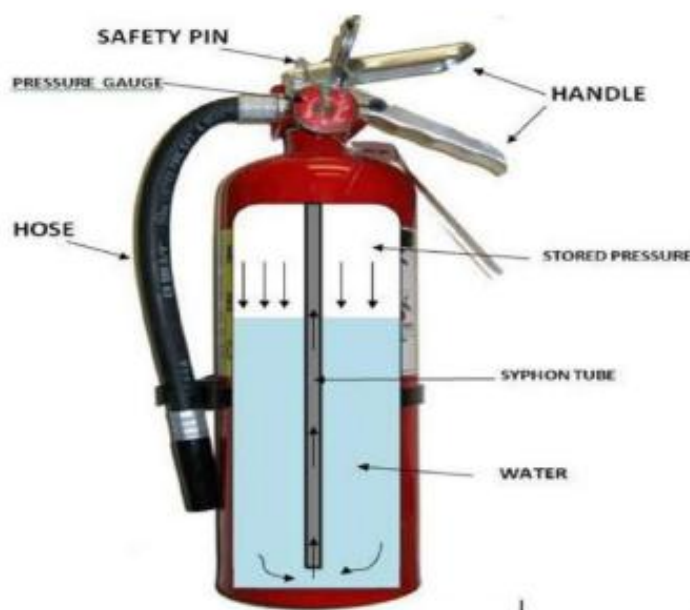
1. Portable fire extinguishers

A fire extinguisher is a handheld active fire protection device typically filled with a dry or wet chemical.

Its purpose is to extinguish or control small fires, often in emergencies. Fire extinguishers are provided for a 'first attack' firefighting measure, generally undertaken by the occupants of the building before the fire service arrives.

It is important that occupants are familiar with which extinguisher type to use on which fire.

Most fires start as a small fire and may be extinguished if the correct type and amount of extinguishing agent is applied whilst the fire is small and controllable. Fire extinguisher locations must be clearly identified. Extinguishers are colour coded according to the extinguishing agent.

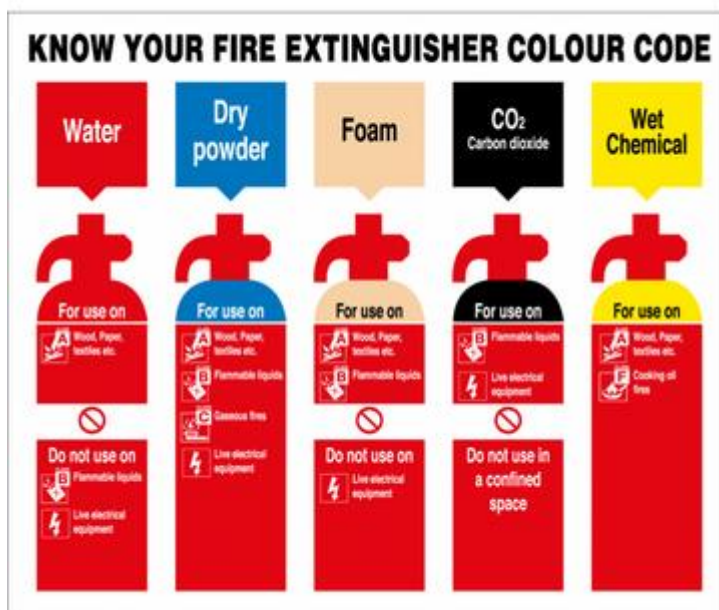


Extinguishing Agent	Color Code	Principle Use
Water	Red	wood and paper fires - not electrical
Foam	Red with cream band	Wood, Paper, flammable liquid, oils and fats fires.
Carbon dioxide	Red with black band	Electrical fires and flammable liquids fires
Dry Chemical	Red with blue band	All fires

Fire Extinguisher Color Codes and Uses:

- **Red (Water)**: For Class A fires (wood, paper, textiles).
- **Cream (Foam)**: Suitable for Class A (solid) and Class B (flammable liquid) fires.

- **Blue (Dry Powder)**: Versatile "ABC" powder, effective for A, B, and C class fires (including electrical).
- **Black (Carbon Dioxide -)**: Ideal for electrical fires and flammable liquids, leaving no residue.
- **Yellow (Wet Chemical)**: Specially designed for Class F/K fires (cooking oils and fats).
- **White (Water Mist)**: Suitable for most fire classes, including electrical. Blackwood Fire Ltd +5



Sand and water buckets are no longer acceptable as a first-aid fire treatment facility

Water is a very good cooling agent, it is inappropriate for some types of fire. It is immiscible with oils and is a conductor of electricity

Chemical foam type of extinguisher - foam is formed by chemical reaction between sodium bicarbonate and aluminum sulphate dissolved in water in the presence of a foaming agent. When the extinguisher is inverted, the chemicals are mixed to create foam under pressure which is forced out of the nozzle.

Carbon dioxide extinguisher - carbon dioxide is pressurized as a liquid inside a cylinder. Striking a knob at the top of the cylinder pierces a disc to release the carbon dioxide which converts to a gas as it depressurizes through the extinguisher nozzle.

It is important to note that fire extinguishers,

- Must be **serviced annually** by certified companies
- Must have **inspection tags**
- Required in all commercial buildings

2. Fire hose reels

A fire hose reel is a hand-operated device used to guide a fire hose from the supply to the fire. Hose reels are firefighting equipment for use as a first-aid measure by building occupants. They should be located where users are least likely to be endangered by the fire, i.e. the staircase landing. The hose most distant from the source of water should be capable of discharging 0.4 l/s at a 6 m distance from the nozzle, when the two most remote hose reels are operating simultaneously. A pressure of 200 kPa is required at the highest reel. If the water main cannot provide this, a break/suction tank and booster pumps should be installed. The tank must have a minimum volume of water of 1. A 50 mm internal diameter supply pipe is adequate for buildings up to 15 m height and a 65 mm internal diameter pipe will be sufficient for buildings greater than this. Fixed or swinging hose reels are located in wall recesses at a height of about 1 m above floor level. They are supplied by a 25 mm internal diameter pipe to 20 or 25 mm internal diameter, reinforced non-kink rubber hose in lengths up to 45 m to cover 800 m² of floor area per installation.

Hose reels are mandatory in:

- Shopping malls
- Universities
- Office buildings

Common issue in Kenya: Hose reels installed but not connected to water supply

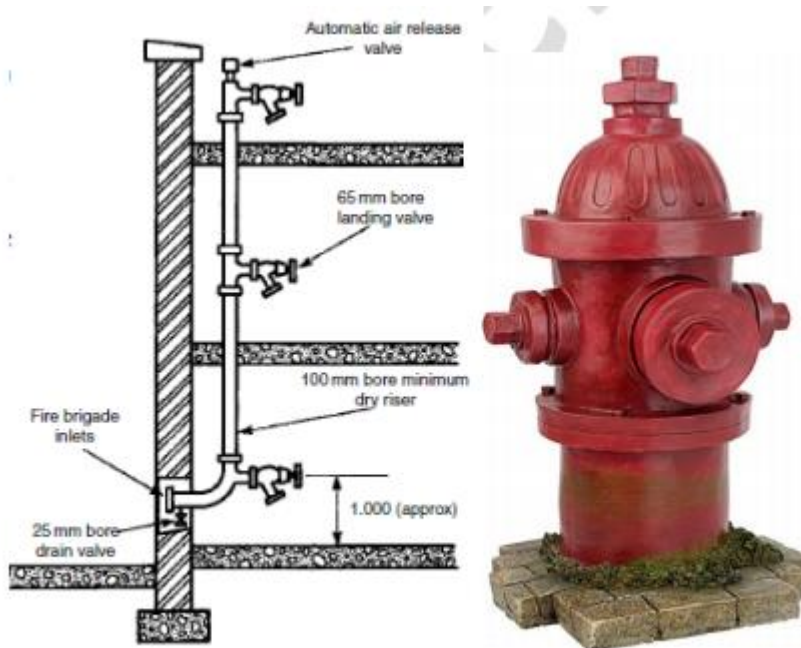


3. Fire Hydrant.

This is a fitting in a street or other public place with a nozzle by which a fire hose may be attached to a water main to provide sufficient emergency water for firefighting.

Fire department connections (fire hydrants) are utilized by the firefighters on the scene to either supply water to the standpipes or supplement the building's already existing water supply. This additional water can be sourced from the fire truck itself or from another hydrant that can tap into water into the building's standpipes (risers). Risers are usually included in the initial creation of the standpipes in order to provide an extra source of water to the building.

Water from a main source via a hydrant is pumped up the building through a riser into each floor. Each floor has a hydrant landing valve.



The riser can be a **dry riser** (charged during emergency) or a **wet riser** (water always present)

Fire hydrant is usually insufficient or non-functional in Kenya, especially in:

- Informal settlements
- Older estates

Fire response handled by Kenya Fire Brigade (County-based)

4. Sprinkler systems.

A sprinkler head is really an automatic (open once only) tap. The sprinkler head is connected to a pressurized water system. When the fire heats up the sprinkler head, it opens at a pre-set temperature, thus allowing pressurized water to be sprayed both down onto the fire and also up to cool the hot smoky layer and the building structure above the fire. This spray also wets combustible material in the vicinity of the fire, making it difficult to ignite, thereby slowing down or preventing fire spread and growth.

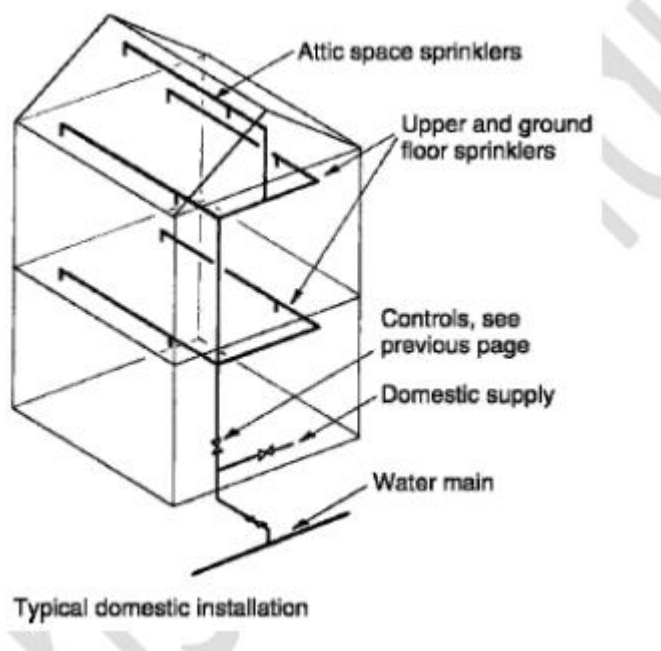
When a sprinkler head operates, the water pressure in the system drops, activating an alarm which often automatically calls the fire service via a telephone connection.

Sprinkler systems are usually installed in high or large buildings and high fire hazard occupancies.

They are mainly used in :

- High-rise buildings in Nairobi CBD
- Hotels
- Warehouses

Limitation: High installation cost → not common in small buildings



Types of sprinkler systems.

The specification of a sprinkler system will depend on the purpose intended for a building, its content, function, occupancy, size and disposition of rooms. Installations to commercial and industrial premises may be of the following type:

- i. **Wet system** –The pipework is permanently charged with water. It is only suitable in premises, where temperatures remain above zero. All pipework is permanently pressure

charged with water and the sprinkler heads usually attach to the underside of the range pipes. Where water is mains supplied, it should be fed from both ends. If the main is under repair on one side, the stop valve and branch pipe can be closed and the sprinkler system supplied from the other branch pipe.

- ii. **Dry system** - installation pipework is permanently charged with compressed air. When a fire fractures a sprinkler head, the compressed air escapes to allow the retained water to displace the differential valve and flow to the broken sprinkler. The rising main is called a dry riser. Due to warm weather, this is not commonly used in Kenya
- iii. **Alternative wet and dry system** – This is essentially a wet system, but due to the slightly slower response time as air precedes water on discharge, the pipework is charged with water for most of the year and only air charged in winter. The maximum number of sprinklers is the same as a dry system.
- iv. **Tail end system:** used in a building with different internal functions, e.g., a mix of office accommodation with an unheated storage facility. The installation differs from an alternative wet and dry system, as most of the pipework is permanently charged with water. Only those pipes in parts of a building exposed to sub-zero temperatures are charged with air and these are designed as additions (tail ends) to a wet system. The wet and tail end parts are separated by a compressed air control valve. As the system is essentially wet, the maximum number of sprinklers may be 1000. The maximum number after a tail end air control valve is 100, with no more than 250 in total on tail end air valves in one installation.
- v. **Pre-action system:** used where there is a possibility that sprinkler heads may be accidentally damaged by tall equipment or plant, e.g. a fork-lift truck, data centers. To avoid unnecessary water damage, the system is dry. If a sprinkler head is damaged, compressed air discharges and an initial alarm is activated. Water will only be supplied to the damaged sprinkler, if a ceiling mounted heat detector senses a temperature rise. The sensor will open a motorized valve on the water supply and effect another alarm. Detectors have a lower temperature rating than the sprinkler, therefore for a 68fC head, the detector will be set at about 60fC. Max. number of sprinklers is 1000.
- vi. **Recycling pre-action system:** a variation of the pre-action system, designed as a damage limiting installation. After sprinklers have subdued a fire, a heat detector responds to a lower temperature and disengages the water supply after a 5-min. delay. If the fire restarts and temperature rise, the detector re-engages a motorized valve on the water supply. Maximum number of sprinklers is 1000.
- vii. **Cycling wet system:** in principle similar to the recycling pre-action system except it is a normal wet system. It functions in conjunction with ceiling heat detectors which will disengage the

water supply within a pre-determined time of the temperature dropping. If the temperature rises, the water supply will be automatically turned on again.

Water Supplies for Sprinkler Systems

There are various sources of water supply that may be used for sprinkler applications.

- a) **Elevated private reservoir** - minimum volume varies between 9 m and 875 m³ depending on the size of installation served.
- b) **Suction tank** -supplied from a water main. Minimum tank volume is between 2.5m³ and 585 m³. A better standard of service may be achieved by combining the suction tank with a pressure tank, a gravity tank or an elevated private reservoir. A pressure tank must have a minimum volume of water between 7 m³ and 23 m. A pressure switch or flow switch automatically engages the pump when the sprinklers open.
- c) **Gravity tank** - usually located on a tower to provide sufficient head or water pressure above the sprinkler installation.
- d) **River or canal** - strainers must be fitted on the lowest part of the suction pipes corresponding with the lowest water level. Duplicate pumps and pipes are required, one diesel and the other electrically powered

5. Drenchers

A drencher fire control system provides a discharge of water over roofs, walls and windows to prevent fire spreading from or to adjacent buildings.

In contrast to sprinklers which are found internally drenchers are normally found on the exterior of a building to protect against a fire from a neighbouring building.

Another difference between sprinklers and drenchers is that all nozzles in the drencher system are activated simultaneously, whereas individual nozzles in a sprinkler system open when activated.

Used to protect **external building surfaces** especially in industrial zones and buildings close together

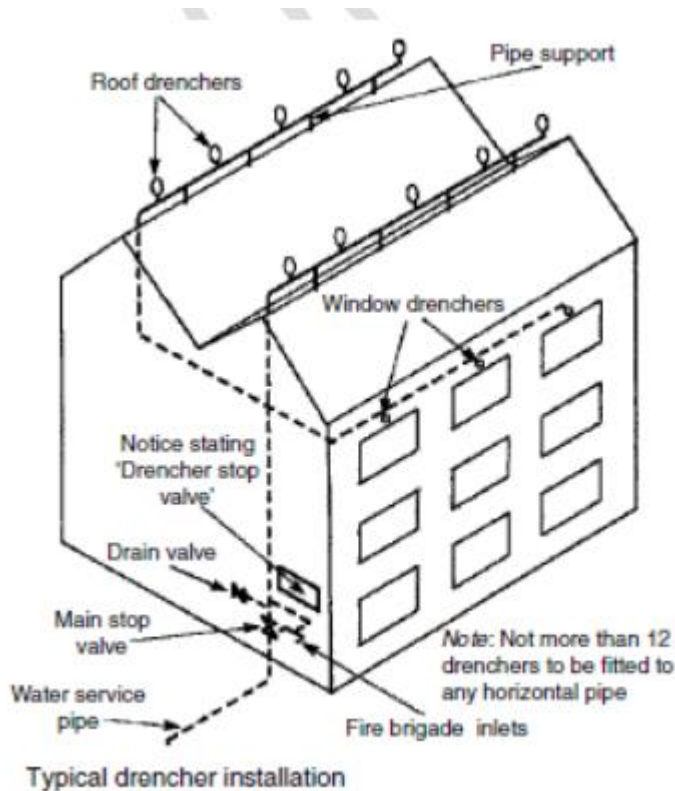
A drencher system consists of the follows components:

- Water source.
- Pump unit.
- Distribution piping.
- Control fittings.
- Alarm device.
- Special nozzles or water head.

The water heads are like those of sprinklers and may be sealed or unsealed

The three main types of drenchers are:

- **Roof drenchers:** Positioned on the roof ridge and throw a curtain of water upwards.
- **Wall or curtain drenchers:** Throw a curtain of water over openings or portions of a building most likely to admit fire.
- **Window drenchers:** Positioned horizontally level with the top of a window so as to protect the opening



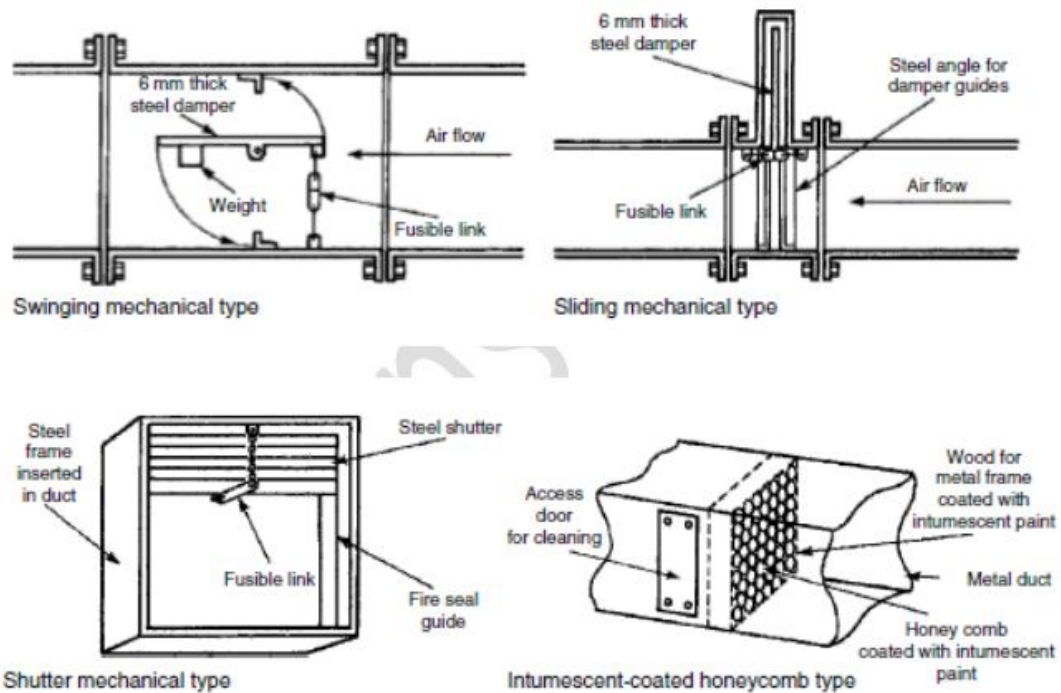
6. Fire Dampers in Ventilation Ductwork

Fire and Smoke Dampers are passive fire protection products designed to close when temperature rise to prevent the spread of fire or smoke inside the ductwork. They are mostly used in ventilation or air-conditioning ductwork.

Dampers should be positioned to maintain continuity of compartmentation by structural division. They can operate automatically by fusible link melting at a pre-determined temperature of about 70°C , to release a steel shutter. An electromagnet may also be used to retain the shutter in the open position. The electromagnet is deactivated to release the shutter by a relay circuit from a fire or smoke detector. The latter is preferable, as a considerable amount of smoke damage can occur before sufficient heat penetrates the ductwork to activate a heat detector or a fusible link.

An intumescent-coated honeycomb damper is an alternative. In the presence of heat, the coating expands to about a hundred times its original volume to form sufficient mass to

impair the movement of fire through the duct. This type of damper has limited fire resistance and is only likely to be specified in low velocity systems.



Dampers are majorly found in: Hospitals

- Airports (e.g., Jomo Kenyatta International Airport)
- High-end commercial buildings

7. Pressurization of Escape Routes

In a building an escape route that is maintained at a higher air pressure than its surroundings; in the event of fire, the higher pressure helps prevent smoke from spreading into the escape route.

A pressurization system is intended to prevent smoke leaking through closed doors into stairs by injecting clean air into the stairwell. The intent is to have the highest pressure in the stairwell and a reducing pressure in the accommodation area to facilitate pedestrian escape route and firefighting access

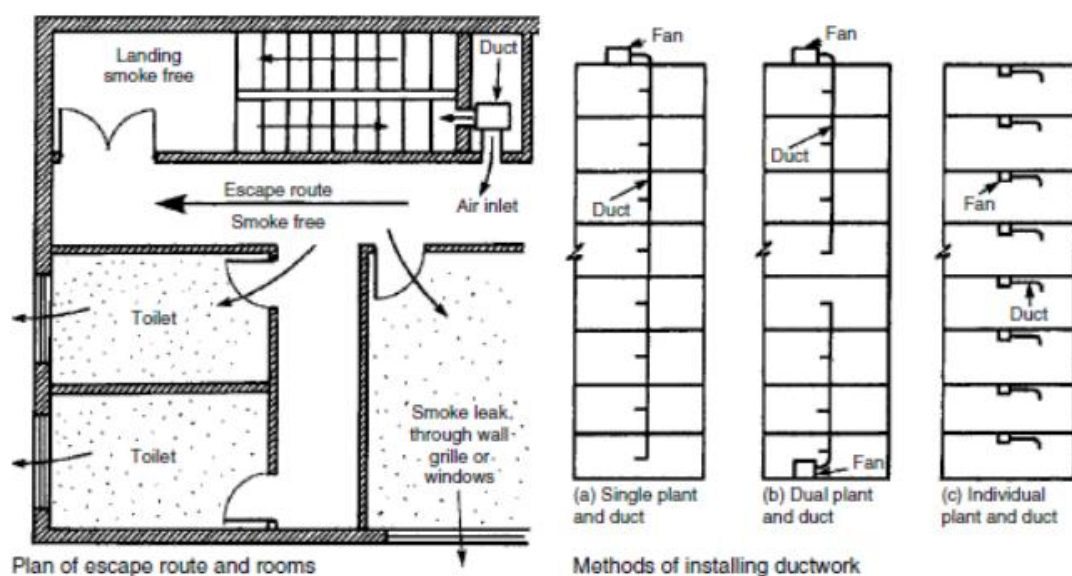
In multi-storey buildings, stairways and lobbies may be air pressurized to clear smoke and provide an unimpeded escape route. The air pressurization is usually between 25 and 50 Pa depending on the building height and degree of exposure. This pressure is insignificant for movement of personnel. A number of pressurization methods may be used:

- Pressurization plant is disengaged, but it is automatically switched on by a smoke or fire detector.
- Pressurization plant runs fully during hours of occupancy as part of the building ventilation system.
- Pressurization plant runs continuously at a reduced capacity and output during the hours of building occupancy, but fire detection automatically brings it up to full output.

It is important to provide openings so that smoke is displaced from the escape routes to the outside air. This can be through purpose made grilles or window vents. Pressurization will help to limit entry of rain and draughts at external openings.

This system is mainly used in high-rise buildings (above ~4–5 floors)

However one of the key Issue is poor maintenance → system fails during fire



8. Fire Detection and Alarms System.

The role of fire detection is to identify a developing fire emergency before making adverse damages. Fire detector would depend on anticipated fire scenario, the type and the use of the building not to mention the occupants and the technicality of the contents. They provide a means to identify a developing fire through automatic and manual methods.

Additionally, they alert occupants to a fire condition and evacuation. The detectors also transmit the signals to all other part of the building or the organization.

A fire detection and alarm system may contain the following components:

- System control unit
- Primary (mains) electrical supply
- Secondary (battery or capacitor stand-by) power supply. An emergency generator could also be used
- Fire Detectors- Smoke Detectors and Heat Detectors,
- Alarm activation devices - manual or automatic
- Alarm indication devices - audible and/or visual
- Remote indication on a building monitoring system
- Control relay via a building management system to effect fire extinguishers and ventilation smoke control actuators.

Types of detectors.

- a) **Human sense:** It is possible for humans to sense a presence. This explains why most detectors are manually activated so that the person who discovers the fire can operate the systems. This detection can also be unreliable since the person may not be there at the beginning of the fire or health complications may make it hard to detect the fire.
- b) **Thermal detectors:** They sense any abnormal rise in temperature. These are periodically spaced on the ceiling. Thermal detectors are easy to install and maintain. However, they may not detect fire at its early stages. They are common in offices, malls
- c) **Smoke detectors:** They are placed along the ceiling and are able to detect fire at its smoldering stage. It is a more modern and more efficient detector. The main advantage of this method is that the fire is detected at its young stage. They are however more expensive to install compared to thermal detectors. Additionally, chances of false alarm are rare. They are common in kitchens
- d) **Flame detectors:** Flame detectors are very expensive and are extremely reliable in a hostile environment. The flame detectors must be directly looking on to the flame which limits its Applicability. They are commonly used in industries

Alarm Output Devices

After the detector senses emergency, it is necessary that it sends the signal. The output system is primarily meant for conveying this notification. **Alarm Systems may include, bells (common), sirens or voice evacuation systems**

The bell is the most common and most familiar sounding device. In areas where a loud signal is needed such as library, and in sensitive architectural structures, horns are ideal.

Thirdly, where a soft alarm tone is preferred, chimes can be used. Good examples of such places entail: theatres and hospital. Speakers are an alternative sounding system and they sound signal that can be reproduced as a recorded message. Large multi-storey buildings will require this kind of Sounders. The main advantage of speakers is that they offer flexibility of emergency addresses to the public.

In areas where the noise level is so high, visual all alerts come hand in hand. In addition, where the occupants have hearing impairment, it is advisable to use visual alarms.

Other output functions are the automatic shutting down of equipment operated by electricity such as computers and air handling fans to contain the smoke and avoid reducing vision in the adjacent building or rooms thus enabling eviction. The system can activate fire extinguishers to put out the fire at its young stage.

Many buildings however **lack functional alarms** and are affected by power outages → need for **backup batteries/generators**

Examples of areas of frequent fire outbreaks in Kenya

Case 1: Gikomba Market Fires

- Cause: Electrical faults + congestion
- Issue: No hydrants, poor access
- There is need for **urban fire planning**

Case 2: Industrial Area Fires (Nairobi)

- Cause: Chemicals, flammable materials
- Lesson: Need for:
 - Sprinklers
 - Hazard classification

Case 3: School Dormitory Fires

- Cause: Arson or electrical faults
- There is need for fire drills and escape routes must be clear

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Conclusion

Fire prevention and control in Kenya is not just a technical requirement—it is a **life safety necessity**.

Effective systems must combine:

- Proper design
- Reliable equipment
- Regular maintenance
- User awareness

Common Challenges in Kenya

- Poor enforcement of fire regulations
- Lack of maintenance culture
- Inadequate training of occupants
- Informal settlements lacking infrastructure
- Corruption in approvals (real-world issue)